

Zofia Chmiel

Weston, CT 06883 • +1 (475) 439-4309 • zofiafchmiel@gmail.com • www.linkedin.com/in/zofia-chmiel

EDUCATION

Duke University

B.S in Mechanical Engineering with Marine Science & Conservation Minor, Class of 2027

Durham, NC

Aug. 2023 - present

RELEVANT COURSEWORK

Dynamics • Thermodynamics • Mechanics of Materials • Linear Algebra • Mechatronics • Controls • Contstructural Theory • Analysis for Design

FIELD ENGINEERING

Hardware Engineering Intern | FaunaLabs - Durham, NC

January 2026 - present

- Contributing to the mechanical redesign of FaunaTag, a biometric wearable for whale health monitoring, using CAD to improve shape, durability, waterproofing, and deployment reliability in harsh marine environments for long-term, field-ready physiological data collection.
- Collaborating with a multidisciplinary team to integrate sensors and optimize form factor for physiological tracking in wildlife applications.
- Applying principles of biomedical device design, waterproofing, and mechanical resilience to enhance the functionality and reliability of FaunaTag for deployment in marine mammal research, supporting ongoing field studies and data collection on whales off the coast of Brazil.

Mechanical Design Lead | Plastic Bioremediation & Toxicity Research - Durham, NC

January 2026 - present

- Designing and prototyping a bioreactor system to test plastic-degrading enzymes and study additive toxicity in lab settings.
- Assisting with integration of biological and mechanical components, including cell culture handling and system fluid control design.
- Applying mechanical design to develop scalable tools addressing plastic bioaccumulation through marine bioremediation systems.

Estimator Intern | Saunders Construction - Denver, CO

May 2025 - August 2025

- Led preconstruction for an ~\$8M structural facility, managing cost estimation, trade coordination, and risk mitigation across civil, structural, and architectural scopes to ensure alignment with budget targets, technical requirements, timelines, and feasibility from concept to pricing.
- Created detailed bid tabs, cost models, and scope matrices using Excel and Bluebeam to support pricing accuracy, subcontractor alignment, and constructability review, contributing to risk mitigation, competitive bidding, and effective decision-making during the preconstruction process.
- Presented value engineering recommendations and collaborated with engineers to align with technical, budget, and schedule requirements.

LEADERSHIP

Project Manager | Durham, NC & Eswatini

August 2024 - September 2025

- Managed the design and planning of a suspended pedestrian bridge and WASH system serving 300+ residents, overseeing stakeholder coordination and ensuring alignment with engineering standards, environmental constraints, and long-term functionality goals.
- Collaborated remotely with engineers, NGOs, and local officials to develop structural plans, perform site feasibility assessments, and ensure safe, sustainable delivery of bridge and WASH infrastructure aligned with technical requirements and community needs.
- Applied systems thinking and risk assessment to improve infrastructure resilience and ensure year-round access to water and transportation.

In Country Project Manager | Kaihura, Uganda

September 2023 - July 2024

- Led an 11-student team to complete an 8-week rural school construction project; managed daily operations, coordinated material logistics, and communicated effectively with local laborers to ensure alignment with design specifications, schedule goals, and safety protocols.
- Adapted construction plans to overcome material shortages and delays while ensuring alignment with project specs and community needs.
- Delivered structural components on time and under budget through agile leadership, team coordination, and on-site problem-solving.

EXTRACURRICULAR INVOLVEMENT

Pratt Tour Guide • Catholic Center Liturgy Director • Duke Theta Tau • Society of Women Engineers • Duke Club Swimming • Kappa Alpha Theta

SKILLS & INTERESTS

Python • MATLAB • Bluebeam • SolidWorks • Autodesk Fusion • Onshape • Polish & Spanish • Microsoft Office • Diving • Folk & Country Music